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Project Title: AXI4-Lite Interface Design using Verilog
Subject: VLSI Design
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Academic Year: 2026

CERTIFICATE

This is to certify that Kottala Naga Venkata Sri Anjani has successfully completed the project titled "AXI4-Lite Interface Design using Verilog" under my guidance as part of the VLSI & Embedded Systems program.

This project is submitted as partial fulfillment of academic requirements.

Teacher Signature:Vaibhav
Date:13-04-2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my instructor for providing guidance and support throughout this project. I also thank my institution for giving me the opportunity to work on this topic.

I am grateful to my parents and friends for their encouragement and support during the completion of this project.

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1. Introduction

AXI (Advanced extensible Interface) is a high-performance communication protocol widely used in modern System-on-Chip (SoC) designs. It is part of the AMBA (Advanced Microcontroller Bus Architecture) developed by ARM.

AXI4-Lite is a simplified version of AXI, mainly used for low-speed control register access. It supports basic read and write transactions without burst transfers, making it easy to implement.

In this project, a simple AXI4-Lite slave interface is designed using Verilog HDL to demonstrate data transfer between modules using handshake signals.

2. Objectives

- **To understand AXI4-Lite protocol**
- **To design a simple AXI interface using Verilog**
- **To implement read and write operations**
- **To simulate and verify data transfer**
- **To understand handshake mechanism (VALID & READY)**

3. Methodology

The project is implemented using Verilog HDL by designing a simple AXI4-Lite slave module.

Steps:

1. **Define input and output signals**
2. **Implement write operation using address and data**
3. **Implement read operation using stored data**
4. **Use handshake signals for communication**
5. **Simulate using ModelSim**

The design uses a small memory block to store and retrieve data.

4. Main Content / Project Details

AXI4-Lite Basics

AXI4-Lite uses separate channels for read and write operations.

Write Channel:

- **Address (AWADDR)**
- **Data (WDATA)**
- **Control signals (VALID, READY)**

Read Channel:

- **Address (ARADDR)**
- **Data (RDATA)**
- **Control signals (VALID, READY)**

Handshake Mechanism

Data transfer happens only when:

- **VALID = 1**
- **READY = 1**

This ensures proper synchronization between sender and receiver.

Verilog Design Overview

The design includes:

- **Memory array**
- **Write logic**
- **Read logic**

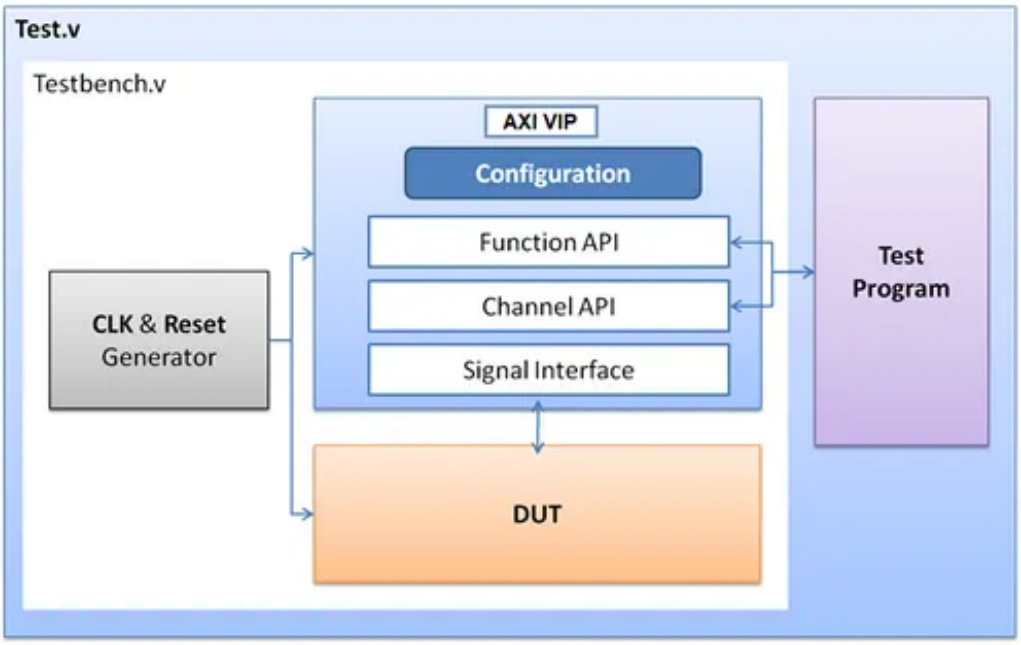
```
module AXI_lite_slave (
 input clk,
    input reset,
    input [3:0] awaddr,
    input awvalid,
    output reg awready,
    input [31:0] wdata,
    input wvalid,
    output reg wready,
    output reg bvalid,
    input bready,
    input [3:0] araddr,
    input arvalid,
    output reg arready,
    output reg [31:0] rdata,
    output reg rvalid,
    input rready)
reg [31:0] memory [0:15];
always @(posedge clk)
begin
    if (reset) begin
```

```
    if (reset) begin
        awready <= 0;
        wready  <= 0;
        bvalid  <= 0;
    end
    else begin
        if (awvalid && wvalid) begin
            memory[awaddr] <= wdata;
            awready <= 1;
            wready  <= 1;
            bvalid  <= 1;
        end
        else begin
            awready <= 0;
            wready  <= 0;
            bvalid  <= 0;
        end
    end
end
always @(posedge clk)
```

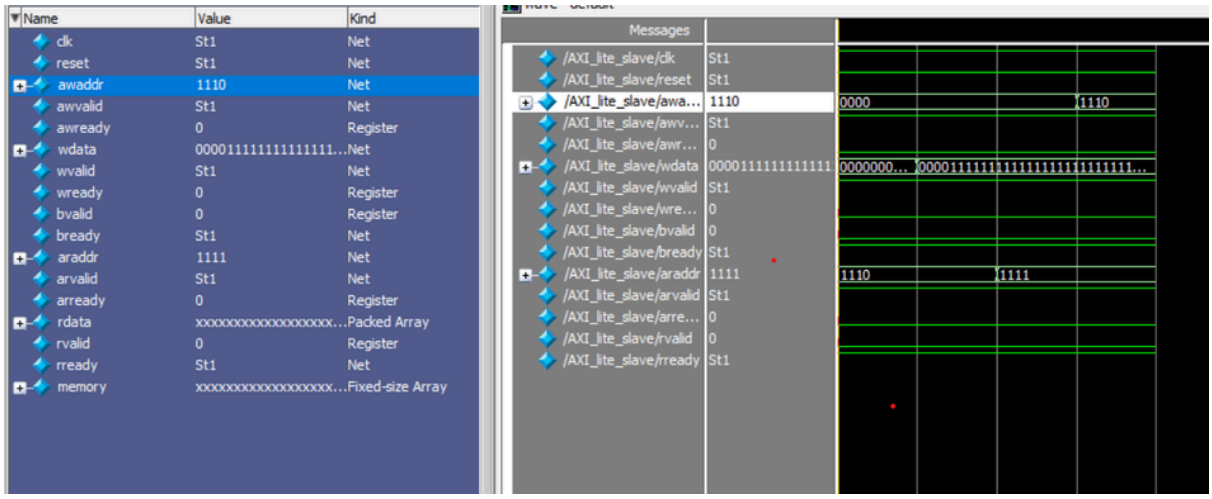
```
always @(posedge clk)
begin
    if (reset) begin
        aready <= 0;
        rvalid  <= 0;
    end
    else begin
        if (arvalid) begin
            rdata  <= memory[araddr];
            aready <= 1;
            rvalid  <= 1;
        end
        else begin
            aready <= 0;
            rvalid  <= 0;
        end
    end
end

endmodule
```

Block Diagram



Simulation Results



Explanation:

- Write operation stores data in memory
- Read operation retrieves same data
- Handshake signals control transfer

5. Observations / Findings

- Data is written successfully when VALID signals are high
- Read operation returns correct stored value
- Handshake mechanism ensures reliable communication
- No data loss observed during simulation

6. Conclusion

The AXI4-Lite interface was successfully designed and simulated using Verilog HDL. The project demonstrates how data transfer occurs between modules using handshake signals.

This project helped in understanding bus protocols and their importance in modern VLSI systems.

7. Bibliography

- **Verilog HDL textbooks**
- **Online resources and tutorials**
- **ARM AMBA AXI documentation**
- **EDA Playground / ModelSim simulations**

SIGNATURE

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